Visualization Plots Guide

Which plot to use at every ML phase — EDA through production monitoring

1 Problem Understanding

	Useful Plot / Why	
get (SalePrice)	Histogram / KDE	See skewness, long tails, multimodality
arget (Titanic)	Count plot / bar chart	Check class imbalance
reme values	Boxplot / violin plot	Detect outliers
s over time	Line plot	See trend, seasonality, drift

2 Missing Data Analysis

	Useful Plot / Why	
missing	Missingness bar plot	Shows which columns need attention
lated to target	Boxplot target vs indicator	Tests if missingness carries signal
co-occur	Missingness heatmap	Patterns across rows/columns
/ group	Grouped bar plot	Example: missing basement by house ty

3 Univariate Feature Analysis

	Useful Plot / Why	
e	Histogram / KDE	Distribution shape
utliers	Boxplot	Detect extreme values
ture	Count plot	Cardinality and rare categories
	Bar plot	Check ordered relationship

4 Feature vs Target Analysis

	Useful Plot / Why	
ression target	Scatter plot	Shows linear/nonlinear relationship
get with noise	Scatter + LOWESS line	Reveals curve/trend

	Useful Plot / Why	
regression target	Boxplot / violin plot	Compares target by category
classification target	Bar plot of target rate	Shows risk/default rate

5 Correlation & Multicollinearity

	Useful Plot / Why	
features	Correlation heatmap	Finds related variables
et relationships	Sorted correlation bar plot	Ranks useful features
nstable	VIF table / heatmap	Detects multicollinearity
predictors	Pairplot on top features	Understand feature clusters

6 Outlier Analysis

	Useful Plot / Why	
e extreme values	Boxplot	Detect feature outliers
target	Scatter plot	See influential observations
s extreme	Residual plot	Detect bad predictions
odel	Actual vs predicted + labels	See if Huber/RANSAC helps

7 Feature Engineering / Nonlinearity

	Useful Plot / Why	
nship	Scatter + smooth curve	Suggests polynomial/spline
eature	Histogram before/after log	Check if log transform helps
ffect target jointly	2D scatter colored by target	Shows interaction
el performs better	Partial dependence plot	Shows learned nonlinear effect

8 Preprocessing Comparison

	Useful Plot / Why	
tation strategies	Boxplot of CV scores	Shows stability across folds
values	Histogram original vs imputed	Ensures realistic values
ng effect	Boxplot before/after scaling	Confirms robust scaling
tes many columns	Feature count bar chart	Tracks dimensionality

9 Algorithm Comparison

	Useful Plot / Why	
odels	Bar plot of RMSE/MAE/R2	RMSE, MAE, R2
models	Bar plot of AUC/F1/Recall	ROC AUC, F1, Recall
odel	Error bars on CV scores	Mean + std deviation
dation gap	Line/bar train vs CV score	Detect overfitting

10 Hyperparameter Tuning

	Useful Plot / Why	
meter	Line plot	Shows best range
meters	Heatmap	Shows interaction
search	Parallel coordinates plot	See good parameter regions
ck	Train vs validation curve	Detect too much complexity

11 Model Evaluation: Regression

	Useful Plot / Why	
on quality	Actual vs predicted plot	Perfect model lies on diagonal
	Residuals vs predicted	Detect bias/nonlinearity
bution	Histogram / KDE of residuals	Check normality/skew
by price range	Residuals vs actual	Detect heteroscedasticity

12 Model Evaluation: Classification

	Useful Plot / Why	
	Confusion matrix	TP, FP, TN, FN
y	ROC curve	General discrimination
ta	Precision-recall curve	Better than ROC for rare positives
lity	Calibration curve	Are probabilities trustworthy?

13 Model Interpretation

	Useful Plot / Why	
	Coefficient plot	Direction and strength
	Feature importance bar plot	Important predictors
mportance	Permutation importance	Model-agnostic

	Useful Plot / Why	
ual prediction	SHAP waterfall/force plot	Local explanation

14 Error Analysis

	Useful Plot / Why	
ctions	Top residual table	Inspect failure cases
gory	Boxplot residual by category	Detect group bias
t range	Residuals vs actual	See if expensive homes are underpredic
n features	Map or grouped residual plot	Detect location bias

15 Production Monitoring

	Useful Plot / Why	
ng values changed	Missing-rate drift bar plot	Data quality monitoring
utions changed	Train vs current histogram	Detect feature drift
els changed	New category count plot	Detect schema drift
ibution changed	Histogram over time	Detect model behavior shift

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Quick Decision Guide

Question	Plot to Use
What does my target look like?	Histogram / count plot
Are there missing values?	Missingness bar / heatmap
Are there outliers?	Boxplot / scatter plot
Is relationship linear?	Scatter + smooth line
Are features correlated?	Correlation heatmap
Which model is best?	CV metric bar plot with error bars
Is model overfitting?	Train vs validation curve
Where does model fail?	Residual plot / confusion matrix
Why did model predict this?	Feature importance / SHAP / PDP
Is production data changing?	Drift plots

Regression Learning Plan

10-phase constructive path from OLS basics to production uncertainty

Core Learning Loop

For every regression topic: **Concept** → **small experiment** → **visual diagnosis** → **model comparison** → **error analysis** → **production concern**

Phase 1: Regression Foundations

Goal: Understand what regression is really optimising.

• Train/test split	◆ Target histogram
• DummyRegressor baseline	◆ Feature vs target scatter
• Linear regression / OLS	◆ Actual vs predicted
• Loss functions: MSE, RMSE, MAE	◆ Residuals vs predicted
• Residuals	◆ Residual histogram
• Bias vs variance	
• Overfitting vs underfitting	

Key question: "Are my errors random, or is the model missing structure?"

Dataset: *California housing or diabetes regression*

Phase 2: Preprocessing & Data Quality

Goal: Data preparation often matters more than model choice.

• Missing values: MCAR, MAR, MNAR	◆ Missingness bar plot
• Simple / KNN / MICE imputation	◆ Missingness heatmap
• Scaling: StandardScaler, RobustScaler	◆ Boxplots for outliers
• Outlier handling	◆ Target by category boxplots
• Categorical encoding	
• Data leakage	

Key question: "Is this missing value actually missing, or does it mean something?"

Dataset: *Ames Housing ID 42165*

Phase 3: Regularisation & Feature Selection

Goal: Understand how models become stable.

• Ridge, Lasso, ElasticNet	◆ Coefficient path plot
• Coefficient shrinkage	◆ CV error vs alpha
• Multicollinearity / VIF	◆ Correlation heatmap
• Coefficient paths	◆ Feature importance plot

Key question: "Which features are useful, redundant, or dangerous?"

Dataset: *Diabetes sklearn / CPU Activity ID 197*

Phase 4: Nonlinearity & Interactions

Goal: Learn when linear models are too simple.

• Polynomial features	◆ Scatter + smooth curve
• Interaction terms	◆ Feature vs target colored by another feature
• Splines	◆ Partial dependence plots
• RBF/kernel features	◆ Residuals before/after nonlinear features
• Log transforms	
• Mutual information	
• Partial dependence plots	

Key question: "Is the relationship additive and linear, or curved and interactive?"

Dataset: *CPU activity ID 197, Ames housing, Insurance charges*

Phase 5: Tree Models & Ensembles

Goal: Learn the strongest tabular regression models.

• Decision trees	◆ Model comparison bar chart
• Random Forest, ExtraTrees	◆ Feature importance
• Gradient Boosting, XGBoost / LightGBM / CatBoost	◆ Permutation importance
• Feature importance	◆ Residuals by feature
• Permutation importance	

Key question: "Does a nonlinear model discover interactions better than manual features?"

Dataset: *Bike Sharing ID 42712, Ames housing, California housing*

Phase 6: Metrics & Target Distributions

Goal: Choose the right metric for the target.

• RMSE, MAE, RMSLE, MAPE	◆ Metric comparison bar chart
• Median absolute error	◆ Residual distribution
• Poisson / Tweedie deviance	
• Pinball / Quantile loss	
• Rule: normal target → RMSE, skewed price → RMSLE, count → Poisson/Tweedie	

Key question: "Does my metric match the real cost of being wrong?"

Dataset: *Housing prices, Bike counts, Insurance claims, Wine quality*

Phase 7: Specialised Regression

Goal: Learn that regression is not one single problem.

• Count regression: Poisson, Negative Binomial, Tweedie	◆ Quantile prediction bands
• Quantile regression	◆ Calibration plots
• Ordinal regression	◆ Spatial residual maps
• Multi-output regression	
• Robust regression	
• Spatial regression	
• Time-aware regression	

Key question: "What type of target am I predicting?"

Dataset: *Abalone, Wine Quality 287, Energy Efficiency 242, California 537*

Phase 8: Validation Strategy

Goal: Learn how not to fool yourself.

• KFold, Repeated KFold, GroupKFold	◆ CV score distribution boxplot
• TimeSeriesSplit, Spatial CV, Nested CV	◆ Train vs CV gap plot
• Rule: random rows → KFold, time data → TimeSeriesSplit, repeated groups → GroupKFold, geographic → spatial split	

Key question: "Does my validation setup match future deployment?"

Dataset: *Any dataset with a temporal or group structure*

Phase 9: Error Analysis

Goal: Stop asking only 'what is the score?' — ask 'where does it fail?'

• Residual analysis	◆ Residuals vs prediction
• Segment-level errors	◆ Top error table
• Error by category / target range	◆ Error distribution
• Largest-error inspection	◆ Residuals by category
• Heteroscedasticity	

Key question: "Which cases does the model systematically misunderstand?"

Dataset: *Ames Housing — analyse by Neighborhood, BldgType*

Phase 10: Uncertainty & Production

Goal: Make regression useful in real systems.

• Prediction intervals	◆ Coverage plot (actual vs PI)
• Quantile regression intervals	◆ Drift monitoring line charts
• Conformal prediction	
• Drift monitoring	
• Missing-rate monitoring	
• Model versioning	
• Retraining triggers	

Key question: "How confident is the model, and is the incoming data still familiar?"

Dataset: *Any dataset — build train / predict / monitor pipeline*

OpenML Dataset Recommendations

The right dataset for each regression learning phase

Full Dataset Map

Learning Phase	Dataset	OpenML ID	Why It Is Useful
Regression basics	Diabetes	44214 / sklearn	Small, clean — OLS, RMSE, MAE, residuals
Full practical regression	Ames Housing	42165	Missing data, categorical, outliers, skewed target
Alternative housing	Ames Housing cleaned	43926	Similar concept, cleaner version
Nonlinearity / basis fn	CPU activity	197	Polynomial features, splines, kernels
Regularisation	Diabetes / CPU	44214, 197	Ridge, Lasso, ElasticNet, coefficient paths
Categorical encoding	Medical charges	42720	Categorical + numeric, skewed cost target
Missing data & robust	Ames Housing	42165	MICE, KNN, MNAR/MAR, Huber regression
Gradient boosting	Bike Sharing	42712	Time features, boosting, RMSLE
Spatial regression	California Housing	537	Latitude/longitude, spatial leakage, spatial CV
Heavy-tailed target	Diamonds / medical	42225, 42720	Log target, Box-Cox, Yeo-Johnson
Count regression	Bike Sharing/Abalone	42712, 1	Count-like targets, Poisson/Tweedie
Quantile regression	Wine quality	287 or 40691	Quantile loss, prediction intervals
Ordinal/rating target	Wine quality/student	287, 40536	Ordered target, ordinal vs regression
Multi-output regression	Energy efficiency	242	Heating + cooling load joint prediction
Large tabular / NNs	HIGGS	23512	Large-scale tabular, MLP/TabNet
Production / drift	Bike / House sales	42712, 42165	Time drift, monitoring, train profile

Top 8 Essential Datasets

1	**Ames Housing** ID 42165	Missing data, encoding, outliers, skewed target, full workflow
2	**Diabetes** ID sklearn	Small, clean — perfect for OLS, residuals, regularisation
3	**CPU Activity** ID 197	Nonlinear features, polynomial, splines, kernels
4	**Bike Sharing** ID 42712	Time features, gradient boosting, RMSLE
5	**California Housing** ID 537	Spatial leakage, lat/lon features, spatial CV
6	**Wine Quality** ID 287	Quantile/ordinal targets, prediction intervals
7	**Energy Efficiency** ID 242	Multi-output regression, joint targets
8	**Diamonds** ID 42225	Heavy-tailed/log-normal target, Box-Cox

Industry ML Engineering Standards

15 concepts that separate production ML from notebook ML

- Required columns, data types, allowed categories
- Value ranges, missing-rate thresholds, duplicate checks
- Tools: pandera, pydantic, Great Expectations

Industry question: "Will this pipeline fail safely if tomorrow's data is different?"

- Target leakage, time leakage, group leakage
- Preprocessing leakage — impute AFTER split
- Feature availability at prediction time — does it exist then?

Industry question: "Would this feature actually exist at prediction time?"

- KFold for random rows, TimeSeriesSplit for time data
- GroupKFold when same user/group appears many times
- Spatial split for geographic data, Nested CV for small datasets

Industry question: "Does validation match deployment?"

- RMSE, MAE, RMSLE, MAPE, sMAPE, Median absolute error
- Quantile / pinball loss, business-weighted error
- Segment-level metrics — overall RMSE can hide group failures

Industry question: "What kind of mistake is expensive?"

- Mean baseline, median baseline, last-value for time data
- Group-average baseline, simple linear model
- Previous production model — always beat it first

Industry question: "Did the model beat a simple, explainable baseline?"

- Pipeline + ColumnTransformer — same code for train and predict

- Saved preprocessors, fixed random seeds, config files
- Versioned artifacts: model.joblib, metrics.json, training_profile.json

Industry question: "Can someone retrain this model six months later?"

- Track parameters, metrics, dataset version, model version
- Track feature set and all plots/artifacts
- Tools: MLflow, Weights & Biases, DVC, Neptune

Industry question: "Which experiment produced this model, and why was it chosen?"

- Grid search → random search → Bayesian (Optuna)
- Always tune after strong baseline — not before
- Nested CV for honest tuning to avoid validation overfitting

Industry question: "Did tuning improve generalisation or just overfit validation?"

- Linear coefficients, feature importance, permutation importance
- Partial dependence plots (PDP), ICE plots
- SHAP values — global bar/beeswarm, local waterfall

Industry question: "Can we explain why the prediction changed?"

- Error by category, geography, price band, customer segment
- Error by time period, error by missingness pattern
- Plots: residuals by segment, MAE by category, top 50 errors

Industry question: "Who or what does the model fail on?"

- Prediction intervals, quantile regression, conformal prediction
- Bootstrap intervals, coverage check, interval width
- Calibration: does predicted 80% PI actually contain 80% of actuals?

Industry question: "Should we trust this prediction, and how uncertain is it?"

- Missing column test, new category test, extreme value test
- Outlier stress test, distribution shift test
- Data corruption test — test what happens with messy production data

Industry question: "What happens when production data is messy?"

- Input drift, prediction drift, missing-rate drift, category drift
- Performance drift — MAE/RMSE when actuals arrive
- Retraining triggers: KS test p-value, missing rate threshold

Industry question: "Is the model still valid today?"

- Model card: intended use, limitations, assumptions, out-of-scope
- Dataset card: collection method, biases, version
- Audit trail, approval workflow — required in regulated industries

Industry question: "Can this model be reviewed, audited, and trusted?"

- Batch prediction vs real-time API prediction
- Scheduled retraining, feature store basics
- Docker basics, CI/CD for ML, Model registry

Industry question: "How does this model actually reach users?"

Ideal Artifacts Per Model

Deliver this folder for every pipeline:

- model.joblib — serialised fitted pipeline
- metrics.json — all evaluation metrics
- cv_results.csv — all cross-validation fold results
- training_profile.json — quantiles for drift monitoring
- schema.json / pandera schema — data contract
- feature_importance.csv — ranked features
- predictions.csv — test set predictions
- error_analysis.csv — largest errors with features
- model_card.md — intended use and limitations
- plots/ — target, missingness, actual_vs_predicted, residuals, comparison

Classification Learning Plan

11-phase path from binary foundations to multi-label and production drift

Core Mental Model

Classification is not only: logistic → SVM → trees → boosting.
It is really: **predict probability** → **choose threshold** → **measure cost** → **explain errors** → **monitor drift**

Learning Loop

- Start with DummyClassifier
- Build simple logistic regression
- Compare probability scores
- Choose decision threshold
- Analyse confusion matrix
- Compare algorithms
- Calibrate probabilities
- Analyse errors by segment
- Save model + metrics + profile
- Monitor data and performance drift

Phase 1: Binary Classification Foundations

Goal: Understand what classification is really doing.

- Binary labels, class balance, DummyClassifier
- Logistic regression, sigmoid, log-odds, log loss
- Decision boundary, regularisation, confusion matrix

Datasets: Titanic, Heart disease, Breast cancer

Key question: "Can my model beat a simple baseline, and what kind of mistakes does it make?"

Phase 2: Metrics & Thresholds

Goal: One of the most important classification phases.

- Accuracy, Precision, Recall, F1, F-beta, Balanced accuracy
- ROC AUC, PR AUC, MCC, Log loss, Brier score
- Threshold tuning — 0.5 is almost never the optimal threshold
- Rule: imbalanced → PR AUC; FN costly → recall; FP costly → precision

Datasets: German credit, Bank marketing, Credit card fraud

Key question: "Is 0.5 really the right threshold? (Usually, no.)"

Phase 3: Data Preparation for Classification

Goal: Learn preprocessing that respects classification structure.

- Missing data, categorical encoding, ordinal encoding
- Rare category handling, class imbalance, leakage prevention
- class_weight='balanced', SMOTE — applied only inside CV folds

Datasets: Titanic, Adult income, German credit, Hepatitis

Key question: "Am I preprocessing in a way that leaks validation information into training?"

Phase 4: Linear Classifiers

Goal: Build intuition for linear decision boundaries.

- Logistic regression, regularised logistic regression
- Linear SVM, SGDClassifier, Naive Bayes
- Coefficients as explanations — log-odds interpretation

Datasets: Breast cancer, Spambase, SMS spam

Key question: "Is the class boundary mostly linear?"

Phase 5: Nonlinear Boundaries

Goal: Learn when linear classifiers are insufficient.

- Kernel SVM (RBF, polynomial), Decision trees, k-NN
- Decision boundary visualisation
- Overfitting with flexible models — tree depth vs train/test score

Datasets: Breast cancer, Sonar, Two-moons synthetic

Key question: "Do I need nonlinear boundaries, or am I just overfitting?"

Phase 6: Trees & Ensembles

Goal: Master the strongest tabular classification models.

- Decision trees, Random Forest, ExtraTrees
- Gradient Boosting, XGBoost / LightGBM / CatBoost
- Feature importance, Permutation importance, SHAP

Datasets: Adult income, Bank marketing, Covertype, Credit-g

Key question: "Does the ensemble improve generalisation or only training performance?"

Phase 7: Class Imbalance

Goal: Deserves its own phase — the most common production problem.

- Why accuracy fails on imbalanced data
- class_weight='balanced', SMOTE, ADASYN, Tomek links
- Balanced Random Forest, threshold tuning
- Evaluate with PR AUC, MCC — not accuracy

Datasets: Credit card fraud, Mammography, Bank marketing

Key question: "Did I improve minority class detection, or only move errors around?"

Phase 8: Probability Calibration

Goal: Very industry-relevant — probabilities must be honest.

- Calibration vs discrimination
- Reliability diagram, Brier score, Expected Calibration Error
- Platt scaling, isotonic calibration, CalibratedClassifierCV

Datasets: Breast cancer, German credit, Bank marketing

Key question: "When the model says 80%, is it correct about 80% of the time?"

Phase 9: Multiclass Classification

Goal: Extend binary classification to K classes.

- Softmax, One-vs-Rest, One-vs-One
- Macro / micro / weighted F1, Top-k accuracy
- Multiclass confusion matrix, per-class ROC AUC

Datasets: MNIST, Fashion-MNIST, Covertype, Wine

Key question: "Which classes are confused with each other, and why?"

Phase 10: Feature Selection

Goal: Find features that are genuinely predictive.

- Mutual information, Chi-square, ANOVA F-test
- L1 logistic regression, RFE, RFECV, SelectFromModel
- Feature importance stability across bootstrap samples

Datasets: Madelon, Gisette, Spambase

Key question: "Which features are genuinely predictive and stable across folds?"

Phase 11: Multi-label & Ordinal Classification

Goal: Special classification structures beyond single labels.

- Multi-label: binary relevance, classifier chains, label powerset
- Evaluation: Hamming loss, Jaccard score, subset accuracy
- Ordinal: Frank-Hall decomposition, Quadratic Weighted Kappa (QWK)
- Datasets: Scene/Yeast (multi-label); Wine quality/ESL (ordinal)

Datasets: Scene ID 312, Yeast, Wine quality 287

Key question: "Does each row have one class, many labels, or an ordered class?"